

Homework 1

MATH 518 - Fall 2025

Due on Friday, September 26, 11:59pm on Crowdmark

Total of points = 75 points

Please submit each question separately on Crowdmark. Unless stated otherwise, k is an algebraically closed field of characteristic 0.

1. (a) (5 points) Show that any prime ideal is radical.
(b) (5 points) Let I be an ideal in $k[x_1, \dots, x_n]$. Show that $V(I) = V(\text{rad}(I))$.
(c) (5 points) Let V be an algebraic set. Show that $I(V)$ is a radical ideal.
2. (a) (5 points) Let I, J be two coprime¹ ideals in a commutative ring R . Show that $IJ = I \cap J$.
(b) (5 points) Write generators of the ideal $I(V) \subset k[x, y]$ where

$$V = V(xy^2, (x-1)(y+1)^2) \subset k^2.$$

3. (10 points) The Zariski closure of a set $W \subset k^n$ is the smallest Zariski closed subset $\overline{W} \subset k^n$ such that $W \subset \overline{W}$. Show that

$$\overline{W} = \bigcap_{\substack{W \subset Y \\ Y \text{ is closed}}} Y = V(I(W)).$$

4. For each of the following algebraic subsets, find the unique decomposition into irreducibles.
 - (a) (5 points) $V(y - x^2) \subset k^2$.
 - (b) (5 points) $V(x^2 + y^2 - 1, x^2 - z^2 - 1) \subset k^3$.
 - (c) (5 points) $V(y^4 - x^2, y^4 - x^2y^2 + xy^2 - x^3) \subset k^2$.
 - (d) (5 points) In each case, are the irreducible components different if we take $k = \mathbb{R}$ (not algebraically closed)?

¹Two ideals are coprime if $I + J = R$.

5. (10 points) Let $V = \{(t, t^2, t^3) \in k^3 \mid t \in k\}$. Find generators of the ideal $I(V)$ and deduce that V is an irreducible algebraic set in k^3 .
6. Prove or provide a counterexample to the following statements.
- (a) (5 points) The intersection of two irreducibles is always irreducible.
 - (b) (5 points) The Zariski topology on k^{n+m} agrees with the product topology² on $k^n \times k^m$.

²The product topology on the product $X_1 \times X_2$ of two topological spaces is the topology with basis the sets of the form $U_1 \times U_2$, with U_i open in X_i . In particular, it makes the projections $p_i: X_1 \times X_2 \rightarrow X_i$ continuous.